NAVIGATOR 1 - ELECTRICAL SCHEMATICS

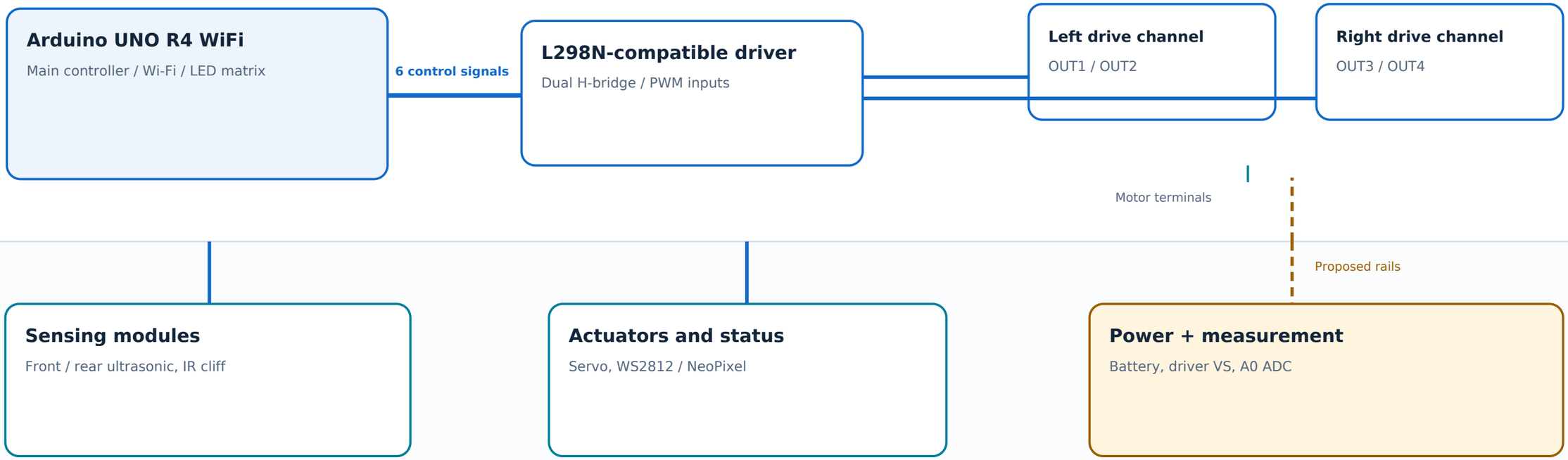
Electrical reconstruction based on main-branch firmware and README. Dashed power nets are design proposals, not confirmed build details.

Firmware-verified signal

Proposed / unverified supply

Check before wiring

FUNCTIONAL SYSTEM TOPOLOGY



SOURCE OF TRUTH

Pin assignments in Navigator-1.ino are used where the README disagrees. Four L298N direction pins differ.

SCHEMATIC STATUS

Control signals are code-derived. Physical motor polarity, battery topology, exact components and power feeds remain unverified.

ENGINEERING REVIEW

Includes electrical cautions, wiring table, source discrepancy report, and firmware safety observations.

GitHub repository

Firmware at reviewed commit

README at reviewed commit

Repository BOM.pdf (binary not inspected)

PROPULSION AND WI-FI MODE SWITCH

Six signal wires are taken directly from Navigator-1.ino. L298N channel output polarity and motor count must be tested physically.



PROPULSION CONNECTION NOTES

M1 / M2 symbols denote driver output channels, not confirmed individual motor quantities. If multiple motors are paralleled, calculate summed stall current against the driver rating.

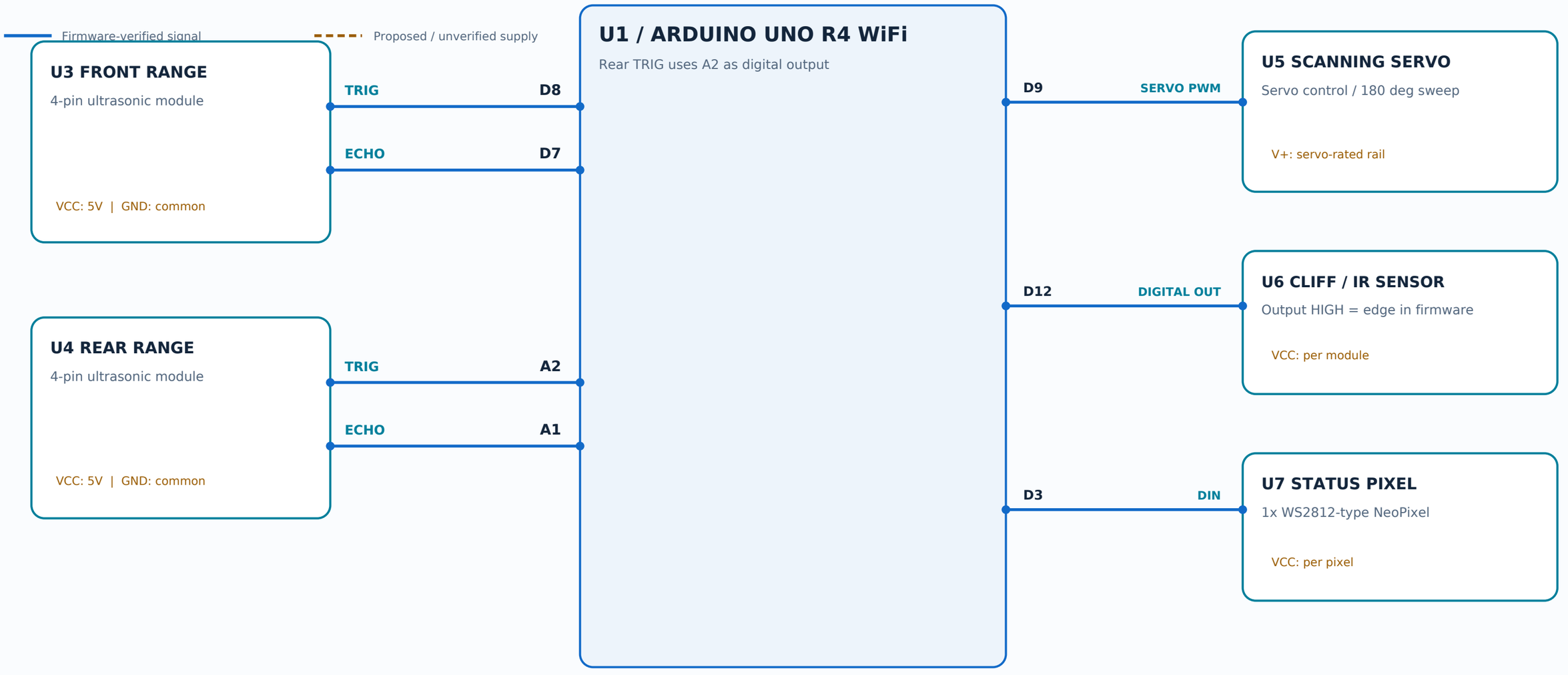
* U2 supply wiring depends on the actual L298N carrier and its 5V regulator jumper. Do not connect two regulator outputs together. The GND return must be common.

SIGNALS VERIFIED

POWER TO VERIFY

DISTANCE, CLIFF, SERVO AND STATUS LED

Module TRIG/ECHO/DIN/signal connections are code-confirmed. VCC and GND power rails are shown only as unverified interface requirements.



UNVERIFIED POWER INTERFACES

U3/U4 typically require a 5 V supply; U5 servo supply must handle startup/stall current; U6 sensor VCC depends on module; U7 requires appropriate LED supply and ground. All share controller ground. Use adequate bulk decoupling and protect the NeoPixel data line as needed.

POWER DISTRIBUTION AND BATTERY SENSE

Design proposal only: repo does not confirm physical battery, regulator, fuse, resistor values, power inlet, or 5 V jumpers.

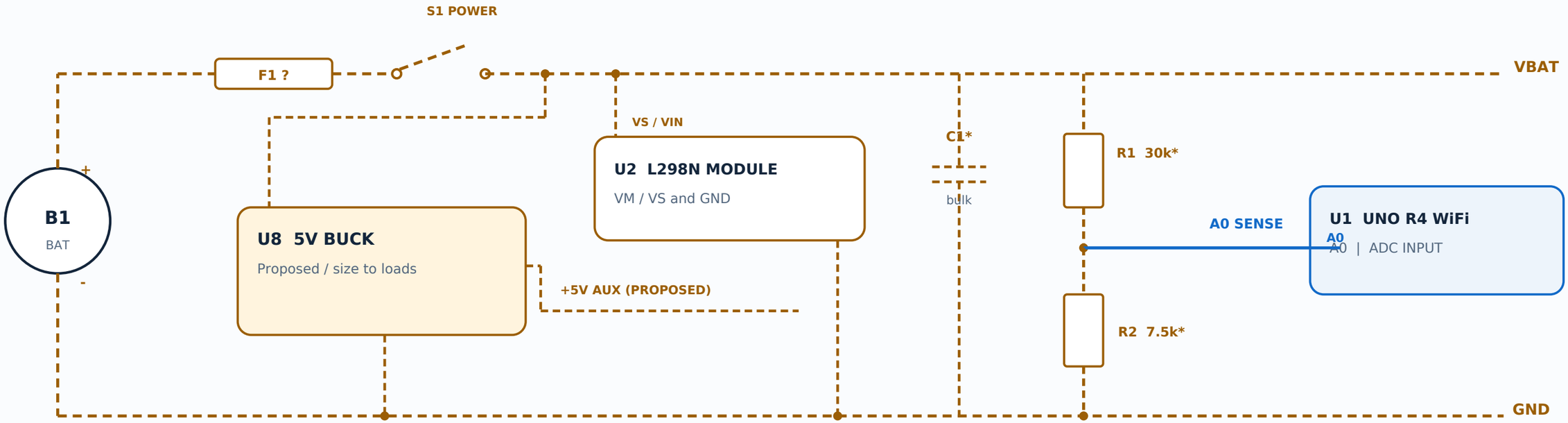
Firmware-verified signal

Proposed / unverified supply

Check before wiring

CAUTION / POWER SHEET IS NOT AN AS-BUILT CIRCUIT

Validate nominal/max voltage, motor stall current, regulator headroom and thermal limits, L298N regulator jumper, UNO R4 supply entry, and exact A0 sensing module before applying power.



MEASUREMENT AND POWER ASSUMPTIONS

* R1 = 30k and R2 = 7.5k are an ILLUSTRATIVE 5:1 divider for a common 0-25 V sensing module, NOT verified bill-of-materials values. A0 must never exceed its ADC input rating. Firmware formula assumes 10-bit ADC counts (0..1023) and an effective 25 V measurement range.

The firmware uses 7.4 V / 6.8 V / 6.4 V thresholds, suggesting but not proving a 2-cell lithium pack. Choose fuse and wire gauges from measured motor stall current. Connect the Arduino through a manufacturer-approved input; do NOT blindly backfeed its 5 V pin or tie regulator outputs together.

Not authorized for assembly until power design and module datasheets have been checked.

PIN ASSIGNMENTS AND ENGINEERING FINDINGS

Code-vs-README discrepancies, measured dependencies, and follow-up verification checklist. Code at commit a945d8b6c969.

BOARD PIN	CODE CONNECTS TO	README CLAIMS	REVIEW
D2	Driver IN4	Driver IN1	**MISMATCH**
D3	NeoPixel DIN	Status LED	**OK**
D4	Wi-Fi mode switch	AP/STA switch	**OK**
D5	Driver ENA (PWM)	Driver ENA	**OK**
D6	Driver ENB (PWM)	Driver ENB	**OK**
D7	Front ultrasonic ECHO	Front ECHO	**OK**
D8	Front ultrasonic TRIG	Front TRIG	**OK**
D9	Scanning servo signal	Servo	**OK**
D10	Driver IN3	Driver IN2	**MISMATCH**
D11	Driver IN2	Driver IN3	**MISMATCH**
D12	IR cliff sensor	IR cliff sensor	**OK**
D13	Driver IN1	Driver IN4	**MISMATCH**
A0	Battery voltage sense	Voltage sensor	**OK**
A1	Rear ultrasonic ECHO	Rear ECHO	**OK**
A2	Rear ultrasonic TRIG	Rear TRIG	**OK**

Reviewed firmware README Repository BOM

Release status: DRAFT - electrically review and bench-test before connecting power or motors.

PRIORITY / SAFETY FINDINGS

- Lost browser heartbeat changes LED status but does not command a motor stop (manual direction can persist).
- Firmware contains plaintext Wi-Fi credentials in a public repository: rotate exposed credentials and remove secrets from source/history.
- D4 uses INPUT with no internal pull-up/down: use a real two-rail SPDT switch; a floating contact is unsafe.

HARDWARE VALIDATION NEEDED

- Exact battery, charger/BMS, fuse, switch rating, L298N module jumper position and power path.
- Motor count and startup/stall current per channel, actual right/left motor polarity and wiring.
- Voltage sensor resistance and calibration; servo 5 V current; IR logic voltage; ground topology.

DOCUMENTATION / FIRMWARE GAPS

- README claims interrupt-level cliff detection; firmware polls D12 and blocks on a HIGH reading.
- Distance sensor timeouts return 100 cm (interpreted as clear), even if a sensor is disconnected.
- turnLeft/turnRight direction comments may not match IN1..IN4 logic: bench-test wheel rotation.
- Source repository includes a BOM.pdf, but binary BOM content was not available for verification.